# BiteFixes Enterprise Architecture Manual

# Volume 11

# Bitey AI Architecture Specification (BAAS)

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# Chapter 1

# Introduction

Bitey is the Intelligent Service Orchestrator of the BiteFixes platform.

Bitey is **not**:

- a chatbot
- a Large Language Model
- a REST API
- a prompt

Bitey is an enterprise AI service responsible for coordinating customer interactions, business knowledge, workflow recommendations, and intelligent assistance.

Large Language Models are replaceable implementation components.

Bitey is the permanent business intelligence layer.

# Chapter 2

# AI Philosophy

The platform follows an **AI-First** architecture.

Business processes are designed assuming AI participation, while preserving human control over critical decisions.

Fundamental principles:

- AI assists.
- Policies decide.
- Workflows execute.
- Humans approve critical operations.
- All AI actions are traceable.

# Chapter 3

# Bitey Architecture

```
                Customer

                   |

                   ▼

          Communication Channel

     (Web / Mobile / WhatsApp / API)

                   |

                   ▼
```

```
                 Bitey Core

         ┌──────────┼──────────┐

         ▼          ▼          ▼

Memory      Intent Engine   Context

         ▼          ▼          ▼

Knowledge   Reasoning   Workflow Advisor

         ▼          ▼          ▼

Policy Engine    Event Bus

               ▼

       Application Services
```

Bitey orchestrates. It does not own business data.

# Chapter 4

# AI Responsibilities

Bitey is responsible for:

- Understanding user requests.
- Detecting language.
- Identifying intent.
- Retrieving knowledge.
- Maintaining conversational context.
- Recommending workflows.
- Suggesting priorities.
- Assisting technicians.
- Explaining decisions.
- Generating summaries.

Bitey is **not** responsible for:

- Persisting business entities.
- Modifying data directly.
- Bypassing business rules.
- Ignoring security policies.

# Chapter 5

# AI Decision Boundaries

Every AI recommendation belongs to one of four levels.

## Level 1 — Informational

Examples:

- FAQ answers.
- Knowledge explanations.
- Device specifications.

Automatically allowed.

## Level 2 — Recommendation

Examples:

- Suggested service.
- Suggested technician.
- Suggested priority.

Requires application confirmation.

## Level 3 — Operational

Examples:

- Create ticket draft.
- Generate estimate draft.
- Suggest workflow transition.

Requires business validation.

### Level 4 — Critical

Examples:

- Financial approval.
- Refund.
- Data deletion.
- Contract modification.

Always requires explicit human approval.

# Chapter 6

# AI Memory System

Bitey uses multiple memory layers.

```
Short-Term Memory

↓

Conversation Memory

↓

Customer Memory

↓

Company Memory
```

```
↓

Knowledge Base

↓

Global Platform Memory
```

Each layer has a different purpose.

## Short-Term Memory

Stores the active conversation context.

Examples:

- Current issue.
- Current device.
- Active workflow.

Lifetime: conversation.

## Customer Memory

Stores long-term customer preferences.

Examples:

- Preferred language.
- Frequently repaired devices.
- Communication preferences.

## Company Memory

Stores tenant-specific knowledge.

Examples:

- Working hours.
- Supported services.
- Warranty policies.
- Internal procedures.

## Global Platform Memory

Stores platform-wide knowledge shared across tenants.

Examples:

- Device models.
- Operating systems.
- Common troubleshooting guides.

Tenant data never leaks into global memory.

# Chapter 7

# Intent Engine

Intent detection converts user language into business actions.

Example:

```
"My printer does not print."

↓

Language Detection

↓

Intent Detection

↓
```

```
printer_support

↓

Suggested Workflow

↓

Knowledge Retrieval
```

The Intent Engine should support multilingual processing and confidence scoring.

# Chapter 8

# Knowledge Retrieval

Knowledge retrieval follows a layered strategy.

1. Company knowledge.
2. Approved service documentation.
3. Device-specific information.
4. Global knowledge.
5. External AI (when permitted).

Priority is always given to company-owned knowledge.

# Chapter 9

# Context Management

Context combines information from multiple sources.

```
Conversation

+
```

```
Customer

+

Device

+

Ticket

+

Knowledge

+

Company Policies

↓

AI Context
```

The context builder ensures responses remain relevant and consistent.

# Chapter 10

# Reasoning Engine

The Reasoning Engine evaluates available information before producing recommendations.

Typical reasoning sequence:

```
User Input

↓

Intent
```

```
↓

Knowledge

↓

Business Rules

↓

Policies

↓

Workflow State

↓

Recommendation
```

Reasoning must remain explainable.

# Chapter 11

# AI Orchestration

Bitey may coordinate multiple AI providers.

Example architecture:

```
Bitey Core

├── Local Models

├── OpenAI

├── Gemini
```

```
├── Future Models
└── Internal ML Services
```

The orchestration layer selects the most appropriate model according to:

- Task type.
- Cost.
- Latency.
- Availability.
- Company policy.

The rest of the platform never depends on a specific AI provider.

# Chapter 12

# AI Skills

Skills are modular capabilities.

Examples:

- Device Diagnosis.
- Knowledge Search.
- Ticket Classification.
- Language Translation.
- Technical Summary.
- Customer Response Generation.
- Estimate Assistance.
- Inventory Recommendation.

New skills can be added without modifying the Bitey Core.

# Chapter 13

# Human Collaboration

Bitey collaborates with humans rather than replacing them.

Examples:

- Drafting responses.
- Suggesting diagnostics.
- Recommending spare parts.
- Summarizing conversations.
- Highlighting risks.

Technicians retain final responsibility for technical decisions.

# Chapter 14

# AI Safety

Safety mechanisms include:

- Policy enforcement.
- Prompt injection resistance.
- Sensitive data filtering.
- Tenant isolation.
- Confidence thresholds.
- Audit logging.

Bitey never accesses information outside the authenticated tenant.

# Chapter 15

# AI Observability

Every AI interaction records:

- Model used.
- Prompt version.
- Knowledge sources.
- Confidence.
- Latency.
- Tokens (if applicable).
- Outcome.
- Feedback.

These metrics support monitoring and continuous improvement.

# Chapter 16

# AI Governance

AI behavior is governed through:

- Business Rules Catalog.
- Policy Engine.
- Workflow Engine.
- Security Policies.
- Audit Requirements.

Model updates do not change governance rules.

# Chapter 17

# AI Evolution

Bitey is designed to evolve independently.

Possible future capabilities include:

- Predictive maintenance.
- Automatic workflow optimization.
- Failure prediction.
- Intelligent scheduling.
- Voice interaction.
- Image-assisted diagnostics.
- Autonomous knowledge generation.

The architecture supports gradual expansion without redesigning existing domains.

# Chapter 18

# Enterprise Vision

Bitey is not an add-on.

It is a first-class architectural component that coordinates intelligent interactions across the entire platform.

Its purpose is to augment human expertise while respecting business policies, workflows, security, and governance.

The long-term vision is to make Bitey the intelligent operating layer that connects every module of BiteFixes without becoming a source of tight coupling.

# Canonical AI Architecture

```
Customer
    │
    ▼
Communication Channels
    │
    ▼
Bitey Core
    │
    ├── Memory
    ├── Intent Engine
    ├── Knowledge Retrieval
    ├── Context Builder
    ├── Reasoning Engine
    ├── AI Skills
    └── Model Orchestrator
             │
             ▼
Policy Engine
             │
             ▼
Workflow Engine
             │
             ▼
Application Services
             │
             ▼
Event Bus
```